

Where the sun does not shine: is sunshine protective against melanoma of the vulva?

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Intermittent sun exposure and sunburn are strong predictors of cutaneous malignant melanoma (CMM). On the other hand, melanomas may arise also in non-sun-exposed areas such as the vulva. However, little is known about a possible relationship between sun exposure and vulvar melanoma. Temporal and latitudinal dependencies of the incidence rates of vulvar melanoma were studied in comparison with those of CMM among Caucasians in Sweden, East Germany, USA and Victoria (Australia). The ratios of vulvar melanoma incidence rates to those of CMM tend to decrease with increasing CMM rates. The incidence rates of CMM have increased with time until recently, while those of vulvar melanoma have either decreased or remain constant. In USA vulvar melanoma incidence rates seem to increase from south to north, while for CMM incidence rates on sun exposed skin areas decrease from south to north. Comparison of latitudinal trends of the incidence rates of vulvar melanomas and CMM show opposite trends. Whenever CMM rates increase, either with time or with decreasing latitude (indicating increased sun exposure) the ratio of vulvar melanoma rates to CMM rates on exposed skin, seem to decrease. Thus, latitudinal trends seem to support the assumption that vulvar melanomas are not generated by UV radiation, and the possibility exists that solar UV radiation, probably via its role in vitamin D photosynthesis in exposed skin, may have a protective effect against vulvar melanoma and should be further investigated.

Melanoma of the vulva: a pigmented lesion is also significant in a non-sun-exposed area.

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The foremost important aetiological factor for malignant melanoma is considered to be sunlight exposure. However, primary lesions are also seen in non-sun-exposed areas. Vulvar melanoma is rare and associated with impaired outcome. Herein, we attempt to increase physicians' awareness for early diagnosis in order to improve prognosis. A 64-year-old female presented with pruritus and irritation at her external genitalia. At examination a pigmented lesion of the vulva 3 cm in diameter was seen. Incisional biopsy revealed melanoma. Clinical examination and imaging studies did not show evidence for metastatic disease. She underwent wide excision of the melanoma with primary wound closure and biopsy of sentinel lymph nodes, which were free of disease. After a follow-up period of 43 months, she remains free of disease. Literature review shows that prognosis is poorer than at other sites, most probably due more advanced disease at diagnosis. This case demonstrates that early diagnosis by physicians aware of this malignancy at this relatively rare site is of crucial importance since it may result in improved outcome, similar to that of melanoma at more common sites.

Mucosal, genital, and unusual clinical variants of melanoma.

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To demonstrate the need for a thorough cutaneous and mucosal examination, we discuss and illustrate the spectrum of mucosal melanomas and unusual clinical variants of melanoma. Although cutaneous areas exposed to sunlight are most vulnerable, melanomas can occur in any site on the skin or mucous

membranes. Pigmented nevi as well as mucosal and labial melanotic macules are lesions that simulate oral mucosal melanomas but are not associated with such a poor prognosis. In contrast, the 5-year survival rate for patients with malignant melanomas of the oral mucosa is only 5%. Similarly, the prognosis is poor for patients who have malignant melanomas of the vulva, vagina, male genitalia, or anorectal area; most patients with such lesions are 50 years of age or older. Subungual and plantar areas are common sites of malignant melanomas, and involvement of the eyelid margin portends a poor prognosis. Other rare variants-desmoplastic, amelanotic, and polypoid malignant melanomas-are associated with local recurrences and metastatic lesions. Early diagnosis is the key to proper treatment and improved survival rate for patients with these unusual variants of melanoma. Increased awareness of the wide variety of clinical features of melanoma should lead to earlier diagnosis.

Primary vulval melanoma and genital lichen sclerosis.

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Lichen sclerosis (LS) is a chronic, inflammatory skin disease with a predilection for the genitalia. Although, the association between squamous cell cancer and genital LS is well established, a link with genital melanoma has not been thoroughly explored. However, we have recently published a case series of penile melanoma where 9/11 (82%) of patients seen over a 10 year period with penile melanoma were retrospectively found to have histological and/or clinical evidence of genital LS on review. The aim of this study was to illuminate further the relationship between vulval melanoma and genital LS by reviewing all the cases managed by our hospital and undertaking a literature review. We identified all the cases with a diagnosis of vulval melanoma over a 16-year period (2006-2022) where histology was available. The clinical notes were retrospectively reviewed, and the histological features of all cases were reassessed by two independent mutually 'blinded' histopathologists. We also performed a literature review of genital LS in patients with vulval melanoma. A total of 11 patients with vulval melanoma were identified for the review. Histopathological review found evidence of genital LS in seven of them (64%). Genital LS was not documented in any of the original histology reports. Clinical notes and letters were available in nine cases. The literature review identified 12 relevant studies with a total of 18 patients. Twelve cases concerned adult women, and six concerned female children. The presence of genital LS in as high as 64% of our vulval melanoma cases might indicate a causative relationship between genital LS and vulval melanoma. The pathogenesis of vulval melanoma remains largely unknown. Although ultraviolet radiation is an important pathogenic factor for cutaneous melanoma, it cannot be a factor in vulval melanoma. While possible mechanisms behind this association remain unclear, it is possible that chronic inflammation from genital LS leads to melanocytic distress and increased mutagenesis.

The Roles of Vitamin D Receptor (VDR) and CD8+ T-Lymphocytes in Acral and Mucosal Melanoma Invasion Depth.

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Acral and mucosal melanomas, the most common sun-shielded site melanoma subtypes in Asia and Indonesia, often yield poor prognoses. The invasion depth reflects their progressivity, and the pathogenesis is influenced by vitamin D receptor (VDR) status and CD8+ T-Lymphocyte amount. This study aims to determine the association between the invasion depth of acral and mucosal melanomas

with their VDR and CD8+ immunoexpression. A cross-sectional observational study was conducted on 60 formalin-fixed paraffin-embedded (FPPE) samples, with equal representation in acral and mucosal melanoma groups from 2017 to 2021. The samples were assessed for the invasion depth and immunoexpression of VDR and CD8+. A chi-square test with an alternative Exact-Fisher analysis was used to determine the association between the variables in both subtype groups. An association between VDR and CD8+ immunoexpression and invasion depth in acral melanoma (p value = 0.0001 and 0.009, respectively) was observed, while only VDR immunoexpression was associated with the invasion depth in mucosal melanoma (p -value = 0.004). Interestingly, no association was found between CD8+ immunoexpression and the invasion depth in mucosal melanoma (p = 0.640). The role of VDR and CD8+ T-lymphocytes are inversely associated with melanoma depth in acral melanoma, while only VDR is associated with melanoma depth in mucosal melanoma.

Ancestry and somatic profile predict acral melanoma origin and prognosis.

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Acral melanoma, which is not ultraviolet (UV)-associated, is the most common type of melanoma in several low- and middle-income countries including Mexico. Latin American samples are significantly underrepresented in global cancer genomics studies, which directly affects patients in these regions as it is known that cancer risk and incidence may be influenced by ancestry and environmental exposures. To address this, we characterise the genome and transcriptome of 123 acral melanoma tumours from 92 Mexican patients, a population notable because of its genetic admixture. Compared with other studies of melanoma, we found fewer frequent mutations in classical driver genes such as BRAF, NRAS or NF1. While most patients had predominantly Amerindian genetic ancestry, those with higher European ancestry had increased frequency of BRAF mutations and a lower median number of structural variants. The tumours with activating BRAF mutations have a transcriptional profile more similar to cutaneous non-volar melanocytes, suggesting that acral melanomas in these patients may arise from a distinct cell of origin compared to other tumours arising in these locations. KIT mutations were found in a subset of these tumours, and quadruple wild-type samples (non BRAF/NRAS/NF1/KIT) differed from mutated samples in their structural genomic profile and overall and recurrence-free survival patterns. Transcriptional profiling defined three expression clusters; these characteristics were associated with recurrence-free and overall survival. We highlight potential novel low-frequency drivers, such as PTPRJ, NF2 and RDH5. Our study enhances knowledge of this understudied disease and underscores the importance of including samples from diverse ancestries in cancer genomics studies.

Nail Unit Melanoma: Uncommon Presentations to Keep in Mind.

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This article presents a case of nail unit melanoma (NUM), highlighting the unusual clinical presentations as those originating from the hyponychium. We discuss how dermoscopy is an essential tool for this pigmented lesion, and how the lack of a standardized guide for NUM underscores the importance of personalized approaches to ensure the best possible outcomes. A 72-year-old woman presented with a melanocytic lesion on her right second finger, characterized by a hyperpigmented macule with irregular borders and a heterogeneous distribution of pigment. Biopsy confirmed acral lentiginous melanoma. Nail unit was exercised with a 5-mm margin toward the hyponychium. NUM is the predominant variant of

malignant melanoma in certain ethnic groups and is influenced by factors such as trauma and chronic inflammation rather than sun exposure. Despite its higher incidence in areas like the great toe and thumb, NUM in the hyponychium is rare. Detection and treatment require thorough examination and individualized surgical approaches. Conservative surgeries may preserve limb function without compromising survival rates. Early detection remains challenging and necessitates attention to patient concerns and potential signs of melanoma.

Mutations in the TP53 gene in human malignant melanomas derived from sun-exposed skin and unexposed mucosal membranes.

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Mutations in the p53 tumour suppressor gene () have been linked to several types of cancer. We therefore investigated whether such mutations occur in malignant melanomas and, if so, whether they are linked to ultraviolet (sun) light exposure. For the first time, mutations in mucosal membranes and adjacent tissues shielded from sunlight were compared with those in cutaneous melanomas from sun-exposed skin. Archival tissues were obtained from 35 patients with a primary melanoma taken from unexposed mucosal areas and from 34 patients with a primary melanoma located in chronically sun-exposed head and neck skin. was characterized by means of polymerase chain reaction amplification and single-strand conformation polymorphism assay followed by nucleotide sequencing. The results showed that 17.6% of the primary cutaneous and 28.6% of the primary mucosal melanomas had point mutations in. Among the cutaneous melanomas, one showed three mutations in exon 7, and one had two mutations in exon 5; the mutation was in the same allele in both cases. One mucosal melanoma had two mutations in exon 7, both in the same allele, and another had two mutations, one in exon 7 and one in intron 6, both in the same allele. C<--T mutations at dipyrimidine sites, considered fingerprints for ultraviolet light-induced mutations, were about equally distributed among patients with melanomas from chronically sun-exposed areas (six out of nine; 67%) and those with melanomas from unexposed mucosal areas and adjacent skin (eight out of 14; 57%). Our data, demonstrating the presence of such mutations even in melanomas from mucosal membranes, clearly suggest that factors other than, or additional to, ultraviolet radiation are operational in the induction of mutations in melanomas.

Environmental factors and the etiology of melanoma.

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In this piece, we have argued that, despite educational programs designed to prevent melanoma, the mortality and incidence rates for the disease have soared steadily. Even on the conventional interpretation of the data, just more than half of the presented cases of melanoma can be explained solely as 'sun-caused.' The conventional interpretation has little to offer by way of explaining the remainder of these cases, some of which, as we discussed earlier, involve melanomas on parts of the body not exposed to the sun. We have argued that the dominant interpretation in which increasing exposures to sunlight account for the increasing incidence of melanomas is a misleading oversimplification. To resolve the melanoma enigma in preventive health terms, it is imperative to acknowledge and to explore the implications of the important associations between the rise in the incidence of melanomas and environmental factors such as our increasing exposure to artificial light,

electromagnetic radiation, and photo-sensitizing chemicals added to many processed foods and a range of medicinal drugs. Our argument has been that these environmental factors contribute to the problem either by altering body chemistry to make it more susceptible to the admittedly harmful effects associated with irregular and excessive exposure to the sun, and/or acting directly, as in the case of electromagnetic radiation, as promoters of melanoma.

Latitude and incidence of ocular melanoma.

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We investigated the associations between latitude and the incidence of two different types of ocular melanoma, external ocular melanoma (exposed to sunlight) and internal melanoma (not exposed to sunlight), separately. Using 1992-2002 data from the Surveillance, Epidemiology, and End Results (SEER) Program of National Cancer Institute, we identified 2142 ocular melanoma cases in non-Hispanic whites, and then regressed the incidences of various types of ocular melanomas with latitude. Our analysis indicated that the higher the latitude (away from the equator, the less sun exposure), the lower the risk of external ocular melanoma (eyelid and conjunctival melanomas) among non-Hispanic whites (P for trend = 0.018). The incidence increased 2.48 fold from 47-48 degrees to 20-22 degrees. This trend is very similar to that of skin melanoma. The incidence of internal ocular melanoma (uveal melanoma) increased significantly with increasing latitudes (the less sun exposure, P for trend < 0.0001), it increased 4.91 fold from 20-22 degrees to 47-48 degrees. The latitudinal patterns of ocular melanomas may reflect the dual effects of sunlight exposure, i.e. a mutagenic effect of direct solar radiation on external ocular melanomas and a protective effect for internal uveal melanoma, which is similar to the sun radiation protective effects for various internal malignant tumors that are not exposed to the sunlight.
